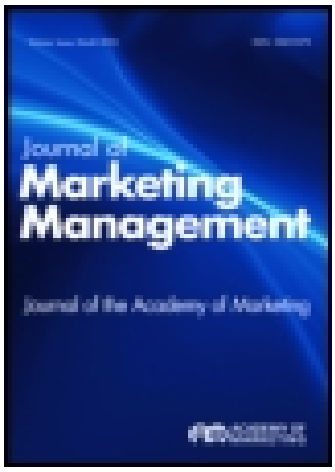
This article was downloaded by: [130.132.123.28]
On: 16 May 2015, At: 01:07
Publisher: Routledge
Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



Journal of Marketing Management
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/rjmm20

Boxing up or Boxed in?: A Short History of the Boston Consulting Group Share/ Growth Matrix

Alan Morrison [a] & Robin Wensley [a]
[a] Warwick Business School
Published online: 06 May 2010.

To cite this article: Alan Morrison & Robin Wensley (1991) Boxing up or Boxed in?: A Short History of the Boston Consulting Group Share/ Growth Matrix, Journal of Marketing Management, 7:2, 105-129, DOI: 10.1080/0267257X.1991.9964145

To link to this article: http://dx.doi.org/10.1080/0267257X.1991.9964145

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at http://www.tandfonline.com/page/terms-and-conditions

Journal of Marketing Management, 1991, 7, 105–129

Alan Morrison and
Robin Wensley

Warwick Business School

Boxing up or Boxed in?: A Short History of the Boston Consulting Group Share/ Growth Matrix

This paper looks critically at the history, mainly from public sources[1], of the development of the Market Share/Market Growth matrix by the Boston Consulting Group and its popularization in both the practitioner and academic domains. The application of the "Boston Box" became a powerful means of simplifying and "boxing up" complex issues of marketing strategy.

However, of particular interest is the question of whether this central technique in any marketing strategy analysis of the seventies or eighties also bred its own form of "marketing myopia" and "boxed in" strategic discussions to a limited set of options and prescriptions. A short survey of marketing lecturers in the UK was also conducted to establish the current state of undergraduate teaching of the "Boston Box" itself, which suggests that there are still considerable areas of concern in terms of the teaching of such techniques.

Downloaded by [] at 01:07 16 May 2015

Introduction

In this paper we trace the development of the Market share/growth matrix (The Boston Box) from its initial development to its widespread adoption. We conclude that the box represents the drawing together and packaging, in an appealing form, of various strands of thought from inside and outside Boston Consulting Group (BCG). The Boston Box met real market needs; particularly senior executives' desire to develop strategic thinking in an increasingly turbulent environment, and to communicate effectively with decentralized subsidiaries. The matrix itself matched the empirically established criteria for the rapid diffusion of any innovation.

We argue that much of the academic criticism has been misplaced. In many cases it treats the box as if it were a "comprehensive" theory of markets and company performance or cites problems which would be true of any comparable technique. However, within UK Business Schools there is a clear lack of consensus on why and how the technique is taught, and on its use and benefits. There is some evidence that the understanding developed in the academic debate and the wisdom of corporate experience, are absent from the teaching in a significant proportion of such schools.

This situation re-inforces concerns as to the extent to which, in practice, the Boston Box has become an approach which "boxes in" strategic thinking and is therefore dysfunctional rather than "boxing up" relevant and useful analysis in an effective manner.

[1] This paper has also benefitted substantially from private communications from Bruce Henderson, Alan Zakon and Seymour Tilles. Any errors or misunderstandings remain, however, the sole responsibility of the authors.

0267–257X/91/020105+25 $03.00/0 © 1991 Academic Press Limited

Antecedents and Development of the Box

Antecedents

In the late 60s, Alan Zakon of Boston Consulting Group (BCG) did some consultancy work for Mead Paper Corporation, particularly related to the development of an acquisition strategy. From this work came the initial development of the notions of portfolio planning and the matrix. These were the methods and tools to be used during an aggressive diversification drive, to solve Mead's problem of developing appropriate strategies for each of their businesses and for "sorting out their losers", from their six product groups and 45 operating divisions (Business Week 1972).

The Boston Consulting Group was started by Bruce Henderson in 1963. From its inception the Group sought to establish itself in the planning area and was considered the pioneer of Business Strategy analysis (Lorange 1975). As early as 1964, Henderson was telling his clients that long range planning should not be equated with 5-year budgets (Henderson 1980): to compete and win required strategic thinking.

Seymour Tilles, who had joined BCG as the third member of staff in July 1964, had set out the main ideas of portfolio planning in an article published in 1966 (Tilles 1966):

—the importance of allocating funds as "the most tangible expression of a company's strategy" (72)

—the need to focus on business generation not cost saving by avoiding traditional "piecemeal" investment by divisions or projects on the basis of cash returns

—the need for more "strategic" funds allocation as competition increased and the business environment became more dynamic

—the need to see the process of funds allocation by product lines as an overall issue of portfolio management.

Elsewhere, during the sixties, leading US corporations such as General Electric (GE) were also looking explicitly at concepts and techniques for strategic planning. This was seen to require careful definitions of product markets, and the term Strategic Business Unit (SBU) was coined at GE to describe a business focused on a particular product-market (Kiechel 1979). Also at GE, work had started in 1960, under the direction of Sid Schoeffler, on the PROM (Profitability Optimisation) project, which had established within 5 years a substantial database with performence for all units within the company.

Alan Zakon, in his Mead project, was interested in the particular nature of the paper business within Mead. It appeared that the paper business was a potential source of growth but that it consumed large quantities of cash: if you wanted to stay in the paper business you needed to acquire a source of cash to fund this growth. Zakon presented his ideas about "cash deficient" and "growth deficient" businesses within Mead and the need for balance between cash generators and cash users to William Wommack of Mead, who was interested in the approach but suggested that Zakon should "dress it up".

Zakon in discussions over lunch with a finance specialist at BCG saw a link between his presentation problem and the more fundamental nature of different forms of financial investment. Broadly speaking he identified: Bonds (where there is

Downloaded by [] at 01:07 16 May 2015

a steady cash flow from the interest and some capital growth); Savings Accounts (where there is no cash flow but compound growth) and Mortgages (where there is a large cash flow but no growth). This led to a three "box" classification to which was added the "Sweepstake"[2] category which was equated with venture capital type opportunities. Hence was born the first form of the Boston Box as illustrated in Figure 1, without any specific axes and solely as a taxonomy. It contained no value judgements about attractive or unattractive quadrants but merely recognized the need for some of balance between the categories, in terms of the cash generated and the re-investment choices. Of course there were some implicit axes but they were not clearly defined: roughly the horizontal axis could be equated with cash generated and the vertical with capital growth.

SAVINGS A/C	SWEEPSTAKE
BOND	MORTGAGE

Figure 1. The Mead Paper Matrix.

The success at Mead led Zakon to present the matrix to various groups within BCG. At the time BCG operated with a very "open" structure with Monday morning meetings of the various staffers in which new ideas and approaches were discussed. Bruce Henderson, who had identified the experience curve effect in his work with the Norton Company in the late 60s and published the general results in a BCG "Perspectives on Experience" in 1968,[3] saw a link between the matrix and an experience curve analysis.

In his 1968 publication, Henderson had introduced the experience curve concept in the following manner:

> "The experience curve concept, developed by the Boston Consulting Group, is relatively new. Unlike the well known 'learning curve' and 'progress function', the experience curve effect is observed to encompass all costs—capital, administration, research, and marketing—and to have transferred impact from one product to another through the process of technological displacement and product evolution".
>
> Henderson 1968

[2] Alan Zakon remembers it as the "Wildcat" category although it does appear that Mead certainly used "Sweepstake".

[3] According to Bruce Henderson, this publication although never publically offered, sold 25 000 copies on the basis of word of mouth advertising and direct mail requests.

Downloaded by [] at 01:07 16 May 2015

Henderson reasoned that the original matrix which effectively considered the sustainable growth rate for the company as a function of its portfolio of cash-generating and cash-using businesses could be directly linked to the experience curve. Because of the experience curve, the company with the highest market share has the highest profit margin, which might be used to finance the growth of other products which were potentially faster growing even though they had started with low market share.[4]

Developments and Derivatives

With the contributions from Henderson and other BCG staff, the Growth/Share matrix began to develop. By 1970 the matrix had developed to the form shown in Figure 2.

The key features and assumptions of portfolio planning and the matrix were (Henderson 1970):

—the notion of a portfolio borrowed from the stockbroking/investment management world. The key elements are a differentiated holding of interests in businesses with variety of potential risk and opportunities. Levels of investment required to secure desired returns are a function of both risks and opportunities;

—the classification of businesses into four portfolio categories. These are quite specifically developed for use in predicting and determining investment requirements and cash flows for each of the company's products. The overall objective is a balance of cash generating and cash using products;

—market share as the horizontal axis, and the explicit assumption of the link between market share and cash generation. Unlike the later versions, the original does not use an explicit relative market share ratio to distinguish between high and low market share products. However this is implicit in Henderson's statement that market leadership distinguishes the two categories.

—Growth of the market, measured in volume (not $) terms. This also has a direct relation to cash needs in that growth requires cash for fixed assets, working capital, spend on R&D, promotion etc. The same is said to be true of gains in market share. In this original publication the break point between high and low growth is not defined.

—Given that no product-market can grow indefinitely, the pay-off from growth comes when growth slows: the pay-off is cash which is generated but cannot sensibly be reinvested in that product.

In the context of strategy planning for products, the matrix can also describe "typical" product life-cycles; these are indicated in the "success" and "disaster" sequences shown in the bottom two diagrams.

Each of the four matrix categories has characteristic cash flow potential and a

[4] This assumption, as Henderson recognized, also avoided the problem of an economic contradiction in that without such other uses for cash, the company with the highest market share should be able to finance the highest future growth rate and hence render the competitive "equilibrium" unstable.

Downloaded by [] at 01:07 16 May 2015

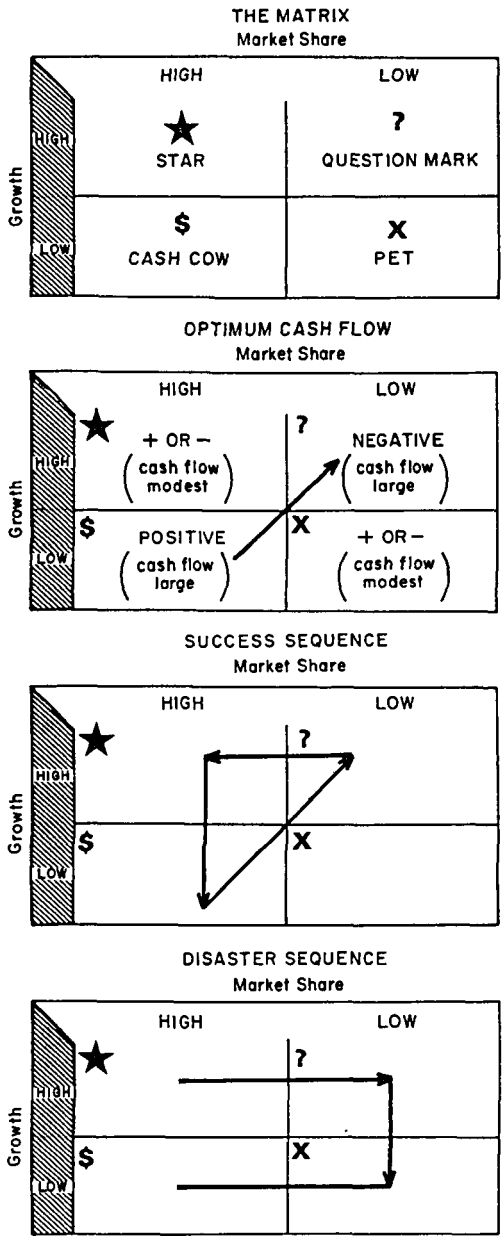


Figure 2. Henderson's Growth/Share Matrix. Source: Henderson, B. (1970) "The Product Portfolio", BCG Perspectives, p. 66.

corresponding strategy prescription in the achievement of the overall goal of a balanced product portfolio:

Stars: high growth and share means significant investment and return. On balance a small negative or positive cash flow. Strategy—invest for the future when market growth slows and the products become;

Cash Cows: market leadership and relatively low costs have been achieved and the

Downloaded by [] at 01:07 16 May 2015

slowing of the market growth requires less investment. Strategy—"harvest" for cash.

Question Marks: products in growth markets (i.e. requiring investment) but not in leadership position. This in turn means lower returns and higher need for investment from headquarters (i.e. significantly negative cash flow). The portfolio cannot support too many of these. Strategy—divest or invest heavily to achieve leadership (Star) status.

Pets: a pet is something which may be nice to have, is a constant, though modest, drain on funds (or small contributor), and is unlikely ever to develop into a star or cash cow. Strategy; divest and cut the losses, unless there are strategic reasons for doing otherwise (e.g. interdependence with other SBUs).

Hence, not only had the labels for the boxes changed, but also some of the logic of the distinctions. By introducing the experience curve into the analysis, the original distinction which was between different forms of investment option, all of which had a role in a portfolio, had become one in which certain forms were explicitly to be preferred. The issue of the costs of gaining a particular position as well as the later benefits had been collapsed into one measure. Indeed even within Mead the original labels continued to be used but the meanings attached to each moved towards the new classification (Business Week 1972).

During the 1970s the BCG matrix underwent additions and modifications, partly based on actual assignments such as the joint consulting work between Henderson and Zakon at American Standard.[5] By 1973 it has assumed the familiar form used in current literature. This is shown in Figure 3.

This version has the now familiar log scale for market share, a percentage scale for market growth, with the break point at 10%, "pets" converted to "dogs" and the relative scales of the businesses indicated by the size of the circles representing them. Henderson viewed investment in growth products as "capital opportunity alternatives", with the 10% figure as the "company investment threshold cut-off rate" (Henderson 1973).

Henderson made a very grand and provocative claim for his matrix;

> "Such a single chart, with a projected position for five years out, is sufficient alone to tell a company's profitability, debt capacity, growth potential and competitive strength".
>
> Henderson 1973, p. 6

In 1977, BCG Director Hedley further spelt out the assumptions underlying the concept and features of the matrix:

—Relative cost position is a fundamental determinant of strategy success (related

[5] With an interesting reflection on current problems with leveraged buy-outs and junk bonds, Henderson recalls that:

> "Its chief executive had built it into a very large and diverse conglomerate as a result of a series of mergers and acquisitions. The company appeared to be a highly successful growth company. However, this had been done with the heavy use of short-term debt in a period that had just preceded a major rise in the rate of inflation which became so severe that the Government put tight controls on the monetary system. They ran out of money."

Downloaded by [] at 01:07 16 May 2015

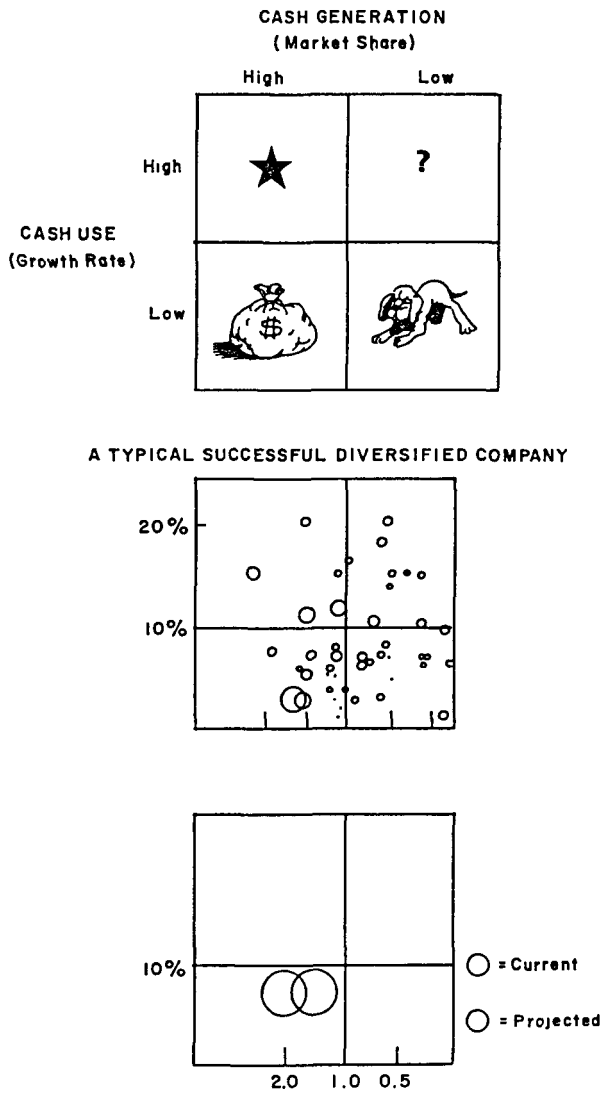


Figure 3. Henderson's Developed Growth/Share Matrix. Source: Henderson, B. (1973) "The Experience Curve Reviewed: IV The Growth Share Matrix or The Product Portfolio", BCG Perspectives, p. 135.

in turn via experience curve to market share), and a worthwhile goal even in an uncertain environment.

—growth is easier in growth markets because of lower levels of competitor reaction.

—The use of a log scale used for share because of "consistency with the geometric progression" of the experience curve effect.

—Hedley also argued that the Growth Share matrix was used partly because the information required to position products on it was *relatively* easy to get and the graph also provided a clear summary of a complex portfolio. He emphasized

Downloaded by [] at 01:07 16 May 2015

that the break point lines were "approximate guides" only. The 10% growth figure was appropriate as an acceptable DCF rate when inflation was low and investment in share was therefore attractive. He also suggested that the Circle diameters for companies can be by sales *or* assets (Hedley 1977).

The BCG matrix is ostensibly a simplifying tool. It selects one parameter, relative market share, as the key indicator of the strength of the SBUs competitive position and one parameter, growth, as indicating the potential and attractiveness of the market. A proliferation of variants followed in the 1970s, as consultants and corporations played tunes on the portfolio planning/matrix theme to suit their own needs and purposes. By 1981, four matrices were in common use and five others also in circulation (Wind & Mahajan 1981). GE produced what is probably the best known alternative. GE were well advanced in corporate planning techniques, at the time of the launch of the BCG matrix, and developed Portfolio planning in parallel with BCG (Schoeffler *et al.* 1974). The GE matrix, as illustrated in Figure 4, involved a nine-box model which used composite parameters of industry attractiveness and business strength. Each of these parameters is constructed from factors selected and weighted by management, as relevant to the particular SBU.

Downloaded by [] at 01:07 16 May 2015

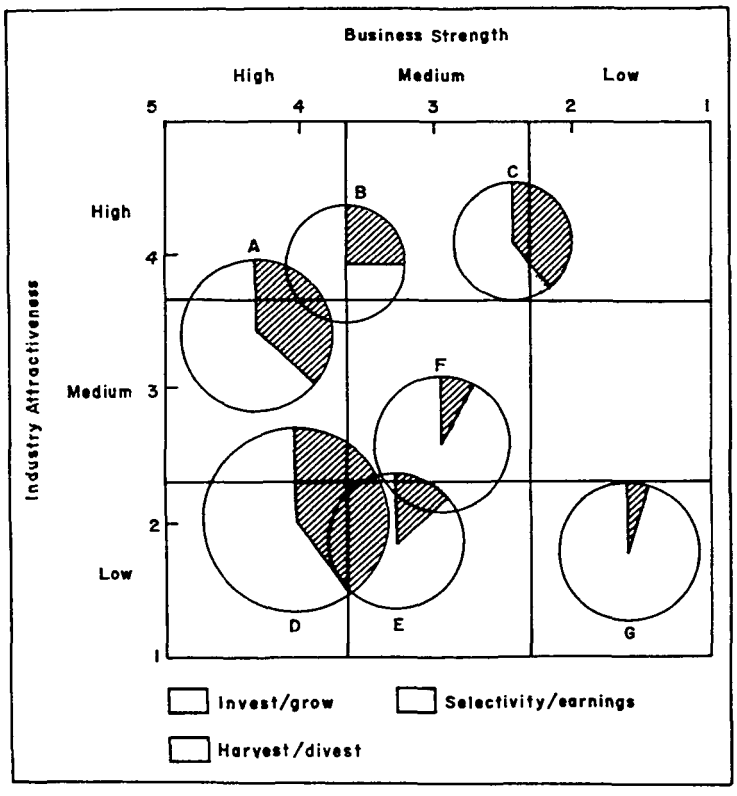


Figure 4. General Electric's Multifactor Portfolio Matrix. Source: Adapted from Strategy Formulation: Analytical Concepts *by Charles W. Hofer and Dan Schendel. By permission West Publishing Company.*

McKinsey also developed a nine box matrix for a similar purpose, at around the same time as BCG. Later, a variety of further matrices were developed. These included BCG's Growth/Gain Matrix (Abell & Hammond 1979), a Competitive Advantage Matrix (Lockridge 1981), Matrix for Market Definition (Day 1981) and a 27 option Share/Strategy Matrix! (Catry & Chevalier 1974).

The Response of US Businesses and its Explanation

Background

Many US businesses in the early and mid 70s were in some difficulties. The uncertainties created by the oil crisis, changing inflation levels and increasing levels of global competition put many companies in a situation of severe economic recession (Hedley 1977). Many companies were in tight circumstances or approaching financial crisis. In these circumstances they were pushed into "belt tightening" cash saving exercises and short-term, undifferentiated, investment criteria.

In parallel many corporations were becoming larger and more diverse. This together with the increased interest in market segmentation (Hewitt 1988), led to the development of autonomous profit centre style management of subsidiaries (Bowman 1974). Thus, in a difficult economic climate, corporate managers wrestled with the problem of how to interact with their divisions (Lorange 1975) and, more importantly, how to influence and co-ordinate their activities.

Corporate strategy was what everyone felt they needed and no one knew how to plan (Day & Wesley 1983). It was the flavour of the 1970s (Kiechel 1979). Corporate managers wanted to "add value" to their collection of businesses, maintain their self esteem and gain the respect of their subsidiary managers. What were the fundamentals of the subsidiaries' business position and environment? (Haspslagh 1982).

A form of communication with subsidiary managers was required which would express the fundamentals of the business and thereby allow real discussion and strategy influence. The previous methods of basing strategies on track record and managerial influence of subsidiaries simply were not working. Undifferentiated instructions, for example to cut 5% from all subsidiary headcounts, were not successful (Haspslagh 1982). It is possible that, at its inception, the BCG matrix was "launched" through Henderson's lines of communication with the Harvard Business School and the corporations connected in various ways to that institution, as well as by direct consultancy work. Whilst this may have ensured a favourable start it did not explain why the ideas spread like wildfire.

The Success of the Ideas

In addition to meeting the important needs of corporate managers in their business situations, various other factors could be hypothesized as contributing to the success of the ideas;

—*"Psychological"*; matrices are very widely used in business and behavioural science text books. The reader is informed "at a glance" of what may take several paragraphs to explain. The concepts have "intuitive appeal" (Day 1977)

Downloaded by [] at 01:07 16 May 2015

and the human "taxonomic urge" fitted well with the urgent need to differentiate between SBUs in strategy terms, particularly in terms of competing claims for limited resources.

—*Fashion;* to quote Wilson and Atkin

> "Businessmen being no less susceptible than the public at large to taking on fashionable theories".
>
> Wilson & Atkin 1976, p. 118

To put the matter more kindly (and rationally), there is a strong potential for the influence of large company orthodoxies over other companies;

> "Well documented and consistent behaviour of successful companies is a strong normative guide . . . practise is only ignored with some folly".
>
> Bowman 1974, p. 48

—*Research;* starting in 1972, the Marketing Science Institute, attached to Harvard Business School, and building on the work started by Sid Schloffler at GE in the earlier PROM project, established a large database of information from a research project called the Profit Impact of Marketing Strategy (PIMS). By 1974 the PIMS study had incorporated 57 large corporations with 620 subsidiary companies. The study confirmed a particularly striking correlation between market share and profitability (Buzzell *et al.* 1975), thereby re-inforcing especially the priority of this item and validating the BCG matrix which used it.

—A further factor was the potential to generate more radical solutions, to allow the company to compete "strategically" rather than "naturally" and eschew solely incremental change (Henderson 1980).

—The existence of formal and informal communications networks. An example of the formal would be directors on the boards of more than one company. It may be no co-incidence, for example, that Vernon Alden was both a director of the Mead Paper and chairman of BCG's parent company at the time of that first successful study.

—The matrix is above all a simplifier. Out of a host of business factors and environmental conditions it selects two as the main focus and start points and shows, simply and vividly, how to apply them to develop strategies. For the users it meant being able to concentrate on collecting a *limited* amount of *specific* information. In Haspslagh's terms it was a "shorthand" (Haspslagh 1982).

Armstrong *et al.* predicted that businesses would be more likely to place a great reliance on the matrix, not just because it was "simple", but because the alternatives (NPV, Risk analysis) were so difficult to work with in terms of getting reliable figures (Armstrong *et al.* 1988).

Corporate strategy was desirable but difficult. Working from "grand theory" at the one extreme and "case studies" at the other was not satisfactory. The BCG matrix is classified (within yet another matrix!) as an analytical approach; as being in "just the right box" by Bowman in his review of corporate strategy approaches (Bowman 1974). Bowman's matrix is shown in Figure 5. Portfolio planning combines methodology with relatively low levels of formality.

The explicit links in the matrix with the ideas of experience curve and product life-

Downloaded by [] at 01:07 16 May 2015

	Less formal	More formal
Practice	Cases	History
Methodology	Analytical approach (BCG)	Management science
Theory	Behavioural	Economics

Figure 5. A Classification of Approaches to the Understanding of Corporate Strategy. Source: Bowman, E. H. (1974) "Epistemology, Corporate Strategy and Academe",* Sloan Management Review, *Winter.

cycle, gave the matrix additional appeal in "drawing together" in an understandable form, the priorities for corporate planning: it certainly provided a means of "boxing up" the strategic planning process. In this respect the matrix had its own "built in" explanation of its logic and purpose. Finally, the publicity generated by BCG, and by the lively debate in the business and academic press, no doubt had a further significant effect on adoption.

Downloaded by [] at 01:07 16 May 2015

Rate of Adoption

By 1972, only 2 years after its public launch, portfolio planning by matrix was being used by over 100 major US companies (Day 1977, Business Week 1972). By 1975 Lorange was able to refer to the matrix as "the common method of corporate planning" and claim that "this type of analysis which is now universally practised" (Lorange 1975, p. 78, 79). In 1977, Day reported that the ideas had "gained wide acceptance among managers of diversified companies" (Day 1977, p. 29).

In 1978 and 1979, Haspslagh carried out research into the adoption of portfolio planning techniques by major US corporations. He concluded that the technique had spread across a wide range of companies by the late '70s and was still being increasingly introduced, although there were some significant differences in uptake from industry to industry. Haspslagh also studied the way the technique was implemented, and discovered that, although some corporations used the method "only" as a top level analytical tool, most used it as a fully integrated part of the management planning process at various levels in the organization.

The matrix was not so much a Eureka discovery or paradigm shift as the adaptation and packaging of existing ideas for its market at that time. Most companies introduced it under conditions of crisis and capital constraint, in situations of uncertainty and competitive pressure (Bowman 1974, Haspslagh 1982, Ansoff 1984). It was perceived as being a tool for communication and influence from corporate centre to its multitude of diversified subsidiaries, allowing the development of "mission statements" and more radical, strategically based, differentiation between SBUs, particularly as a framework for resource allocation. Its focus was on the corporation's prime area of concern; the market and their competitive position within it:

"The primary objectives of corporations, implicit in the

> initial conceptualisation of BCG, are growth & profitability".
>
> Hax & Majluf 1983, p. 50

Many corporate officers felt that in portfolio planning they had found the means to add value to their businesses and thereby maintain their authority and self esteem.

In his extensive studies of factors affecting the rates of diffusion of innovation, Rogers identifies a number of factors which are empirically positively correlated with rapid innovation (Rogers 1983). These are:

—*Relative advantage over alternatives*; the BCG matrix was perceived as focusing on the right things; market, growth, competitive position. Its visual impact was also a huge advantage.

—*Compatibility*; it was consistent with the "religion of growth", and the perceived importance of market share.

—*Simplicity*; was the great virtue of the BCG matrix. This enhanced its communicability, both formal and informal, and relative ease of use.

—*Trialability*; it was possible to use it in limited form (corporate aid) for trial purposes, before commitment to full integration within planning process.

—*Observability*; the rate of adoption by major companies was highly observable, through the press, academic writings, consultancy and PIMS—many communication channels were available and used.

—*Change Agents Efforts*; BCG and other consultancies made great efforts to introduce portfolio planning because of self interest. It could be argued that corporate officers were also well motivated to introduce the technique and, more importantly, well placed in terms of authority and influence to ensure its adoption.

Simple casual explanations are inappropriate for the explanation of the dissemination of innovative ideas. Whilst the foregoing analysis undoubtedly gives an overstated impression of consistency of the circumstances and responses of businesses, the factors described do "fit" together to form a mosaic picture of the period, which provides a plausible narrative understanding of the success of the BCG ideas.

The Academic Evaluation

Whilst the US business community was revelling in their "new" discovery, the academic community was busy dissecting the BCG and other matrices, and evaluating their failings and shortcomings. The matrix had become the new orthodoxy; the target up there to be shot at, as academics sought to enhance their reputations by producing incisive criticism and offering, variously, their own warnings, variants and alternatives. During the 1970s the rapid spread of the ideas does not appear to have been seriously impeded by these critiques; perhaps for BCG, all publicity was good publicity. The perceived real value to the consumer overrode the "technical" shortcomings, in the absence of any better alternative which could fulfil the same needs. In Appendix 1 we detail the nature of the various academic criticisms and consider their overall validity.

Downloaded by [] at 01:07 16 May 2015

Defining the Breakthrough

In general we would conclude that though the criticisms of the technique are real, they are, in general, either difficulties which beset any strategic planning exercise or cautionary tales against over-simplified or thoughtless use. Set against this are the potential benefits of the technique. Haspslagh's study of the use of the technique by major US corporations concluded that its adoption could properly be considered a breakthrough rather than a fad. A breakthrough which gave "permanent added capacity for strategic control" (Haspslagh 1982, p. 73).

The technique helps with the dilemma between centralization and autonomy; i.e. mission influenced by the centre, autonomy for operational control. It gives a start point or springboard for strategic thinking, particularly in companies where this is new. It can be seen therefore as *initiating management development*; as providing a "simple" and conceptually appealing framework *from which to start* out on the long hard road of strategic planning. At Mead, for example, a key change was perceived from the voting of resources as a measure of the manager to the measure of the market; the start of management re-education (Business Week 1972). As a start point, it has some real virtues. It focuses sensibly on markets, their definition, growth and profit potential, on awareness of competitors (as threat rather than "fact of life") and their strategies and relative position, and on the role of sales volume and cumulative experience in making possible (though not generating) cost reduction which in turn enhances profitability and competitiveness. Mead regarded the shift in focus from the profit centres (and their fixed assets) to the market as *the* most important contribution. This is echoed in a quote from one of their SBU managers, worried about the market opportunities for his product:

> "Frankly we wanted to get out, but having the newest plant, we didn't think you would let us".
>
> Business Week 1972, p. 125

It gives a demonstration of what strategy or mission statements look like, and has measurably caused a shift in the perspectives of operating managers to longer term priorities (Haspslagh 1982). It introduces the idea of the role of strategy in resource allocation. The technique can (and perhaps should) be customized to meet the individual market circumstances of the user. The process of customizing is itself a strategic thinking process (Wind & Mahajan 1981) and one by which the managers can come "to own" the technique.

In these cases portfolio planning can represent the

> "creation of a pattern of influence that corresponds to the nature of the business, its competitive position and its strategic mission."
>
> Haspslagh 1982, p. 63

From the top management's point of view, the operating managers of the corporation must make the best use of its assets. This in turn requires forms of thinking which allow for the possibility of radical changes, of real differentiation between units. If a technique such as BCG's offers an attention-grabbing idea, which initiates the dialogue between centre and operating unit, and begins the long process of management education and embodiment of strategic thinking in the life of the organization, then it is undoubtedly worthwhile. How it matures will depend in the

Downloaded by [] at 01:07 16 May 2015

main on top management's commitment and their awareness of the techniques, pitfalls and limitations if applied blindly as a piece of "recipe knowledge". To quote Alan Zakon, the matrix

> . . . "made a major contribution to strategic thought . . . today it is misused and over exposed. It can be a helpful tool, but it can also be misleading, or worse, a straitjacket".
>
> Lorenz 1981

Diagnosis or Prescription?

The growth in global GDP is slowing. International competition is intensifying, and major changes in global market share are expected. For example, one commentator expects Japan's share of total World GDP to grow from 12·5% to 22% between now and the year 2000. In this context, a technique which takes as its startpoint market definitions, growth rates and competitive standing cannot be all bad. Perhaps we should follow Peter's & Waterman's advice and be "close to the customers". In this case the BCG customers were the major American corporations. They had real needs which made them rush and buy, and continue to use, the product. This cannot be wholly disregarded even in the light of the welter of academic criticisms. BCG themselves took on board some of the criticisms levelled at the matrix in the 70s. They recognized that strategies in pursuit of market share and low-cost position alone met unexpected difficulties in terms of effective competition from segment specialists and multiple competitors with scale economies. The new BCG "matrix for the 80s", shown in Figure 6, took more account of the structure of competition. It recognized that share leadership/cost reduction strategy works best in the "Volume" section (Hax & Majluf 1983). The influence of the new Industrial Organisation Economics and particularly the Michael Porter orthodoxy emanating from Harvard is evident in this kind of approach.[6]

It is a real worry that the original matrix is so seductively simple, and the temptations and risk of using it "off the shelf" are real. If the market is simply taken as the trade associaton figures, the competition as the trade association members, the cost savings as materializing automatically from experience, and (probably worst of all) the SBU as the existing operating unit (thereby forestalling possible discussions of restructuring), the use of the technique would be at best unhelpful and at worst positively damaging.

Wise users, such as Mead, recognized the risks. Mead leaders McSwinney and Wommack did not accept BCG proposals immediately;

> . . . "this is a billion dollar company and if you start fooling around with theories you can bomb the hell out of it".
>
> Business Week 1972, p. 125

[6] It is also noteworthy that the newer version of the BCG matrix has hardly stood the test of time in the same way as the original. Perhaps Michael Porter's widely used "5 Forces" diagram (Porter 1980) had pre-empted this particular approach!

Downloaded by [] at 01:07 16 May 2015

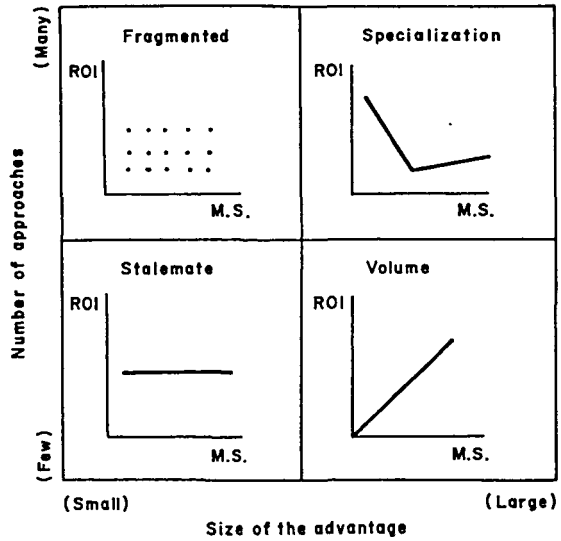


***Figure 6.** Underlying relationships between* ROI *and market share in the new BCG matrix. Source: Hax, A. and Majluf, N. S. (1983) "The Use of the Growth Share Matrix in Strategic Planning"*, Interfaces, *13, No. 1, February.*

In the last analysis no planning technique alone will guarantee good strategies (Lockridge 1981). There will be other preconditions of success. In many instances strategy will be "emergent"; managers in practice, may not connect strategy to investment projects and may use "Boston Box" logic as the rhetoric for post factum justification of actions or decisions, especially if their personal incentives are linked to short-term profitability. In Kiechel's view, synergy and portfolio planning have both failed to make the best use of corporate assets, and that this is the job of the CEO, under pressure from corporate raiders and other corporations who can generate more value for stockholders from their possession of the assets (Kiechel 1988). In parallel, Hewitt sees global shortage of leaders who can achieve success as the key scarce resource. This is now recognized by some major corporations. For example, GE have as their No. 1 corporate objective ". . . to attract, retain and develop the best leaders" (Hewitt 1988).

The overall conclusion may be then that the BCG matrix is a useful tool in initiating corporate planning and strategic change, in organizations where this skill is underdeveloped, and where the main pitfalls of its use can be avoided by wise central management. The main danger in use depends on whether:

> "the positioning of the concept is as a diagnostic aid or a prescriptive guide".
>
> Armstrong *et al.* 1988, p. 3

Eighteen years after its inception, it may have served its purpose in the majority of large corporations; in the management development and strategy planning process. As Townsend has it; if it's management orthodoxy it must be out of date, as it confers no advantage (Townsend 1970). Those who now use it may be "boxed in" in terms of restrictive assumptions about both the nature of market and competitive dynamics.

Downloaded by [] at 01:07 16 May 2015

The Teaching of the Matrix in UK Business Schools

This section describes a survey conducted in the first term of the academic year 1988–89. In order to have a sample of reasonable size and comparability it was decided to survey all the institutions which carried out Undergraduate Degree teaching in business, management and related subjects. The survey, using a short questionnaire, covered 26 universities and 35 other higher education bodies (mainly polytechnics), which run Bachelors degrees. The latter are referred to as polytechinics in the text. The actual respondent targets were the staff teaching on these courses, who taught the matrix as part of their syllabus.

Completed questionnaires were received from 16 universities (61%) and 18 polytechnics (51%). Three universities and one polytechnic returned two completed forms, where the matrix was taught on more than one course. This brought the total number of teachers completing the questionnaires to 38. In every case numbers and percentages given relate to respondents, unless otherwise stated.

Downloaded by [] at 01:07 16 May 2015

How Widely Taught?

One hundred percent of the sample institutions taught the matrix on one or more courses. This confirms expectations and the status of the BCG matrix as an "orthodoxy". In the majority of cases (55%) the matrix was taught as part of a Business Studies degree course. In the remainder of cases, the topic was also taught on a wide range of other degrees, particularly in the University setting.

The matrix is most often taught as part of a Marketing or Marketing Strategy course (74%). The other main arenas are Business Policy (29%) and Business or Management Strategy (24%). Many respondents cite more than one course on which the matrix is taught. In almost all cases the Matrix was taught as part of a wider topic on the course: Portfolio Planning (26%), Marketing Strategy (21%), Strategic Planning (18%), were the only topics well represented, numerically.

The Main Points Covered

One of the most striking features of answers in this section was the number of respondents (50%) who did not mention the techniques, problems, criticisms and weaknesses at all in their list of points covered. A further 29% made only a general reference or statement about "problems" or "weaknesses" without specifying what these were. This tends to support the view that there may be a lag or discrepancy between the current state of the academic debate and what the students are taught: the possibility arises that significant numbers of students are being taught this appealing and seductive technique as simple "recipe knowledge".

Some respondents were clearly aware of the problems of the matrix, as the somewhat higher level of "prompted" response on problems for organizational usage shows. However, if they regard these as significant learning points, it is still difficult to understand why they are so little mentioned in this section. In terms of specific weaknesses identified as teaching points the most frequently mentioned problems were:

Weakness of assumptions	4
Too simplistic	4
Practical problems/measurements	4

Interestingly, what is in some ways the most fundamental problem, the definition of the market was mentioned by only one respondent!

In terms of the teaching points, the most frequently mentioned were:

	No.	%
Description/logic/rationale	23	60
Cash use/balance	14	37
Strategic alternatives/choice	12	31
Product life-cycle	9	24
Relation to other concepts/models	7	18
Practical usage	5	13
Experience curve	5	13

Although it is difficult to interpret what is included in the Description/logic/rationale category, it is still somewhat surprising to find mention at low frequency of such fundamental items as importance of market share (3%), experience curve benefits (13%), how to calculate scales/positions (5%) and the low level of reference to case studies (8%) and practical usage (13%).

Downloaded by [] at 01:07 16 May 2015

Other Matrices

The response showed that other matrices are widely taught (89%), but that no individual matrix is as widely taught as BCG. One forms the impression that BCG is "the" matrix taught, with one or two others added for comparison. By far the most commonly taught of the other matrices are GE (50%) and the Shell Directional Policy Matrix (47%).

The Reasons for Teaching the Matrix/Insights for Students

There was a considerable lack of consensus in the responses given to this question. Over 30 different specific reasons were given, however the most frequently cited reasons were as follows:

—Product/SBU market; positions/assessments/opportunities	37%
—Widely used/part of business language	29%
—Resource allocation/cash balance	24%
—To show problems/care required in use	24%
—To enhance strategic capacity/choice making	21%
—Tool for corporate analysis	18%

Some minority answers are also noteworthy. One respondent answered in this section that the technique gives the students little insight because of their lack of experience. One respondent replied that it was taught for the very good reason that it had "always been done this way". This may be a real explanation in many cases. An exciting new paradigm gets on to the syllabus, once it becomes fashionable; but

such items do not get removed from syllabuses so quickly. This is part of the process whereby text books and teachers perpetuate orthodoxies, in many cases long after they have become discredited, dangerous or straitjackets.

It is worth noting that of the three most cited reasons for use, two are a repetition of the stated purposes of the Matrix (i.e. Product/market assessments/positions and Cash balance use) and one is a statement that it is taught because it, or its terms, are believed to be widely used.

How Widely Used in Industry?

This was, of course, an attempt to gauge, not the extent to which the matrix is used in Organizations, but the perceptions and beliefs of teachers about its use. The question was open ended. Responses can be classified as follows:

Usage	%
Not at all	8
Little used	37
Sometimes used	18
Widely used (general) / Widely used (large organizations only)	24
Unclassifiable	5
Don't know	16

Note: In some cases two answers have been given and entered.

Once again, a clear picture/consensus fails to emerge from the responses. Forty-five percent of respondents believed that the matrix was used little or not at all in industry. Whilst a total of 24% believed that the matrix was used widely, either in general, or in large corporations. Several respondents made the point that the matrix was quite widely used by consultants, and in company training sessions. Of the nine respondents who believed that the matrix was widely used, in general or in large organizations, five had given this as one of their reasons for teaching the matrix.

Perceived Benefits/Problems for Organizations

Respondents were prompted on the "problem" issues. Some of the more interesting findings from this section are summarized below:

Benefit	%
Portfolio appreciation/whole business review	26
Simplicity/appeal	24
Cash allocation/balance	16
Encourages/starts strategic thinking	16
SBU/product assessment	13

The most frequently cited problems were:

Downloaded by [] at 01:07 16 May 2015

Problem	%
Measurement/access to and reliability of information	26
Misleading/untrue/not realistic	26
Simplistic/limited	26
Product-market definition	18

It is interesting to note that a similar (significant) percentage (24%) give "simplicity" as a benefit and give "simplistic" (26%) as a problem, however in only two cases do the same respondents refer to both "simplicity" as a benefit and "simplistic" as a problem. This clearly shows the "double edged" nature of such tools. It is also interesting to see that some 16% of the sample would be in agreement with the main conclusions of this paper, that the principal benefit is in encouraging/starting strategic thinking. It is also rather surprising to see a technique is being taught by the 26% of respondents who believe that the matrix is misleading, untrue or unrealistic.

The State of the Teaching?

A stern sceptic may reach the following conclusion on the above evidence:

The matrix is taught universally as part of business degree courses. The way it is taught is highly variable, and significantly lacking in proper critical review. There is little consensus on the reasons for offering this to students, and such reasons as are given are not well thought through. Taken as a group, those teaching the matrix have no clear idea whether the technique is used in industry, or what real benefits/problems it brings. It is taught because it is on the syllabus, and it got on the syllabus because it was famed. It is a badly taught, outmoded and discredited orthodoxy, which is seductive and dangerous for our young managers of tomorrow, and which is likely to remain on syllabuses for years to come.

Downloaded by [] at 01:07 16 May 2015

Conclusion

Our conclusion would be rather that the Boston Box was a technique for a season rather than one "for all seasons". Its development and adoption however indicate yet again the power of simple and effective presentation particularly when it both addresses some of the concerns of the audience and is supported by both some theory and some evidence. Our greatest concern would be that in our teaching we fail to reflect this and continue to teach what is seen as ideas in good currency after others have made a more balanced and critical evaluation. Certainly Bruce Henderson has somewhat changed his view in terms of his claims for the matrix, from his original assertion that the single chart was sufficient alone to tell "a company's profitability, debt capacity, growth potential and competitive strength", he would now see it more as:

> "a milestone on the search for insight into business system dynamics, but certainly not the end of the road."

It might also be appropriate for the teaching of the Boston Box, if we too focused attention more on the underlying issues of competitive market dynamics and saw

the development and success of the matrix as *inter alia* a case history in successful innovation and diffusion of a particular analytical framework.

Appendix 1

An Evaluation of the Academic Criticisms of the Boston Box

The criticism levelled against portfolio planning in general and the market share/growth matrix specifically, can usefully be considered in a number of broad categories.

Focus/Scope of Technique Some writers have criticized the matrix for not being comprehensive (Abell & Hammond 1979). Other writers have picked on specific features of the matrix, and claimed that the focus was too narrow: on investment decisions (Hewitt 1988), market share (Catry & Chevalier 1974, Day & Wensley 1983, Wind & Claycamp 1976) and supply-side economics (Wensley 1981).

Assumptions/Evidence Criticisms related to a simple focus on internal funding (Wensley 1981, Hax & Majluf 1983), the value of investing in growth markets (Wensley 1981, Wind & Mahajan 1981), to competitive value of market share (Day 1977), the link between market share and cash flow (Abell & Hammond 1979), particularly in the case of "dogs" (Hambrick *et al.* 1982).

Definitions and Classifications Wind, Mahajan & Swire (1981) looked at the Boston matrix classifications of 15 Fortune 500 companies, using four different share and four different growth definitions and showed that only four out of the 15 companies were consistently classified. Similar points were made by Goold (1981b) and Majaro (1977). More specific issues include the 10% growth rate (Kotler 1984), poor market definition (Fruhan 1972, Lorange 1975, Wilson & Atkins 1976).

Political Processes The process of fixing strategies and allocating resources is not a "neutral" one. A unit manager is seen as a loser if the business features prominently in the "dog box" (Haspslagh 1982), even if there are suitable strategies for improvement (Goold 1981a). The labels can be "a vulgar and destructive vocabulary" (Hax & Majluf 1983, p. 55).

Implementation Haspslagh (1982) observed that implementation frequently occurred in crisis conditions when managers were more receptive to the redistributions of power this entailed. Even then linking strategy decisions to project capital expenditure decisions was particularly difficult; in this survey only 14% of companies did this.

Strategic Statements The classical "mission statements" of the BCG matrix may still leave a lot to be desired. Its detractors pointed out that it may show the corporation where it is and where to go, but not how to get there (Lorenz 1981). In particular, "harvesting" cash cows (Lorange 1975), recognizing interdependencies (Hax & Majluf 1983), and generally establishing operational decisions (Wind & Claycamp 1976) can prove very difficult.

An Evaluation of the BCG Matrix

We argue that three key questions need to be answered:

1. Is strategic thinking and decision making an asset for large businesses in general?

Downloaded by [] at 01:07 16 May 2015

2. Is the BCG technique a contribution to strategic thinking?
3. Does the technique allow a corporation to do strategic thinking in the right sort of way?

Generally, there is a strong consensus in answering yes to question one. Ansoff, for instance, states, albeit somewhat tautologically, that research supports the view that strategic planning leads to superior performance, when properly implemented (Ansoff 1984). More detailed empirical research generally comes down on the side of the value of planning. In terms of question two, whilst it would be far-fetched to claim that BCG's offering is a fully comprehensive, all embracing manual for strategic planning, it is difficult not to concede that it is offered as, and has been widely accepted as, a contribution to this activity. Question three is therefore the key question. Any technique or tool for strategy planning is likely to be dangerous or unproductive if used in a mindless or simplistic manner. We will therefore re-phrase this question. Does the BCG technique offer a useful start point for strategic thinking? What are the potential risks and benefits involved? Can the technique be used for net gain?

Downloaded by [] at 01:07 16 May 2015

The Criticisms

Before weighing each category of criticism covered in the last section, it is important to distinguish between the notions of validity and relevance of criticisms. It is possible to criticize a Mini for not being able to do 0–60 mph in 6 seconds, or carry six people in comfort. These criticisms may be *valid* in relation to specific (valued) criteria i.e. speed and space, but not relevant to the small economy car market.

The same appears to be true in relation to many of the criticisms of the BCG techniques. This means that some criticisms are levelled at the technique as if it were a comprehensive economic theory or company model. To be fair, this is understandable in the light of Henderson's original extravagant claims for the matrix. Such criticisms (understandably) focus on shortcomings of the technique in relation to the normal requirements of the "scientific" community for theories such as consistency, clear empirically proven assumptions, and comprehensiveness.

It is more sensible to think of BCG's technique as a tool rather than a theory. A hammer may behave in accordance with Newton's laws of motion. However, when judging a hammer it is not relevant to ask "how well does it exemplify or demonstrate these laws", but "is it a good tool for banging in the range of nails we encounter?" It is not relevant that it will not turn screws or cut wood. Other important questions are "is it easy to use?", "can you hurt your fingers easily?" etc. With these provisos in mind we now turn to consider the various types of criticism discussed earlier.

Focus In getting corporate executives started on the difficult business of strategic planning, the simplicity and narrowness of focus of the technique is its outstanding virtue, i.e. from the great mass of possible concepts and variables, it makes a choice and sets some priorities. If the first hammer to be marketed had three handles, six knobs and four switches for operation, it would undoubtedly not have been a best seller, and the first wooden huts would have been slower in the building.

However the criticisms of narrow focus are, for the most part valid (but not fatal),

and carry a warning not to mistake the hammer for the wooden huts or blueprints for it, but to regard it as a useful tool which is supplied with some helpful tips on hut construction.

Assumptions Once again, BCG's technique has key features to maximize customer appeal. The principal assumptions of giving priority to growth (of markets or share) resonates with the values of the age; growth is wonderful, growth is desirable, growth is a measure of the success and importance of a corporation *and its executives*. For the most part, the academic criticisms point to the limitations resulting from the fact that the correlation of ROI with market share is much less than 100%. Though this would presumably apply to any other comparable technique. Markets are not all the same; gaining share may be difficult, costly, undesirable and not necessarily easier in growth markets. Experience doesn't guarantee cost advantage, neither does scale.

A hammer is good for banging in nails, but not with your eyes closed! Any strategic planning technique needs to be thoughtfully applied to be beneficial. It is true that Henderson's original paper (Henderson 1970) does convey an impression of instant "plug in" answers on strategy and investment. This would be a fair criticism, although this pamphlet should in fairness be considered more of a "sampler", an advertisement for his product. The "real" product is the more detailed strategy consultancy work, and this is unlikely to be as "simple" as the brochure. If there were no more to the product than what is in the brochure, then Henderson would have had nothing left to sell!

In the particular assumption of internal funding, Henderson's original paper does mention leverage and the stockmarket's role in "controlling it", and therefore recognize connections between share price, gearing and profitability. Similarly, Hax & Majluf point out that the key assumption is really "a belief that ultimately any external debt will have to be matched by internal cash flow" (Hax & Majluf 1983).

Definitions/classifications These criticisms are, in practice, a catalogue of the difficulties of working with the technique. They show the "arbitrary" nature of the scales, the criteria and the variability of the resultant classifications. Since these classifications are the lynch pin of the subsequent development of mission statements and broad investment orientations, these are potentially both valid and relevant criticisms.

Here is where the thoughtfulness of the applications needs to be in evidence. The identification (or formulation) of strategically meaningful markets and product/SBUs cannot be taken as "given" but probably *the* major determinant of the success or failure of the planning process. The iterative "double-fitting" process between product/SBU and the market is the most difficult but important step in strategy formulation. Spotting the potential "substitutes" is particularly difficult.

Henderson himself was well aware of these difficulties and admitted that errors had been made where companies improperly identified product-markets, i.e. where the share measured was not of the *relevant* market (Henderson 1973). Armstrong's unpublished research claims to show that the use of Boston Boxes to guide investment decisions will often give the "wrong decision" by NPV "normal profit maximization criteria" (Armstrong *et al.* 1988). However this is based on a short class exercise by graduates, and merely serves as a further warning against simple-minded application.

Therefore, these criticisms need to be heeded. They need not, however, require the technique to be discarded, but to use it with great care and above all to resist the

Downloaded by [] at 01:07 16 May 2015

temptation (which it must be admitted the technique generates) simply to stick existing businesses on a chart using conventional (easy to obtain) market definitions and information. The chance to ask broader questions about competition from other sectors/product types and about the need for SBU restructuring must not be missed, or much of the potential benefits will be lost, as the structure needs to be established to encourage the distinctive behaviour which the firm needs (Ansoff 1984).

Political Processes/Implementation In reality, these two issues will be inextricably intertwined, as the integration of the technique at various levels in the corporation is the outcome of a political process. Whilst the difficulties identified in this area are undoubtedly real they are no greater than the difficulties which any corporate planning "formula" would face. The methods and care in implementation are critical if the technique is to become more than a new set of cynically manipulated game rules to replace the old. Some guidelines have been identified from empirical studies to guide successful implementation (Haspslagh 1982).

As Ansoff points out, strategic planning is neither "natural" nor "welcome" (Ansoff 1984). Management education and differentiated reward systems are therefore a key condition of implementation, as they would be for any strategic plan which changed performance standards. From "basic" ROI, EPS or ROS, *any* changes can be perceived as threat and/or opportunity by unit managers. However, education may not be sufficient. It will need to be recognized that a new strategy for an SBU may require the hiring of new managers with different capabilities to the old and detailed plotting to anticipate, minimize and overcome resistance to the necessary changes (Ansoff 1984). The real risk is that the strategically crucial part of the exercise, the definitions of markets, SBUs and share, may be distorted by subsidiary managers for reasons of pride or self interest, and that this may be undetectable to the "distant" corporate executives. However, the dialogue between broad views (e.g. in seeing shared factors—R&D, distribution etc.) and the narrow one (e.g. in seeing fine detail and profitable sectors) can be helpful.

Where an SBU is designated a fruitful investment area there is still the major difficulty of deciding which projects to invest in. Wolman (1988), based on research by Marsh *et al.* (1988) indicated that the corporate officers could influence investment decisions in terms of questioning basic assumptions, underwriting risk corporately to which the SBU manager may be averse in terms of his own resources, setting deadlines and testing the critical element of commitment in the managers. Such actions can be guided, to some extent, by the kinds of mission statements which the technique can be used to generate.

Strategic Statements Whilst it is true that simple formulae such as "harvest", "milk for cash" etc. cannot be unequivocally and straightforwardly operationalized, the technique could, if used with care and discipline, generate mission statements which serve as a clearer guideline for the SBU unit managers. The power of these "strategic thrust" formulations, in relation to market share, is demonstrated by the fact that many other approaches retain the basic BCG categories (Hax & Majluf 1983).

Again, this problem is a generic one of any strategic type of prescription, and not specific to the Boston Group matrix, though it could be argued that as this matrix is in some ways the "simplest" and most "specific" of the family it is more likely to be translatable into operational deeds. The longer term issue is the establishment of a pattern of influence from Centre to unit. There is no one "best way" to do this (Goold & Campbell 1987). The more doubt there is about the mission and direction of the business unit, the stronger needs to be the centre's involvement in the planning

Downloaded by [] at 01:07 16 May 2015

process. The clearer the operational objectives become and the more mature the product becomes, the more emphasis can be given to control via financial criteria.

References

Abell, D. F. and Hammond, J. S. (1979) *Strategic Market Planning*, New York, Prentice Hall.**

Ansoff, H. I. (1984) *Implanting Strategic Management*, New York, Prentice Hall.**

Armstrong, J. S., Brodie, R. J. and Ringbeck, J. (1988) "The Value of Portfolio Planning Methods for Strategic Marketing Decisions: Experimental Results", Working Paper, August.

Bowman, E. H. (1974) "Epistemology, Corporate Strategy and Academe", *Sloan Management Review*, Winter, pp. 35–50.

Business Week (1972) "Mead's Technique to Sort out the Losers", March 11, pp. 124–130.

Buzzell, R. D., Gale, B. T. and Sultan, R. G. M. (1975) "Market Share—A Key to Profitability", *Harvard Business Review*, **53**, Jan/Feb, pp. 97–106.

Catry, B. and Chevalier, M. (1974) "Market Share Strategy and The Product Life Cycle", *Journal of Marketing*, **38**, No. 4, October, pp. 29–34.

Day, G. S. (1977) "Diagnosing the Product Portfolio", *Journal of Marketing*, April, 29–38.

Day, G. S. (1981) "Strategic Market Analysis and Definition: An Integrated Approach", *Strategic Management Journal*, July–September, pp. 29–38.

Day, G. S. and Wensley, R. (1983) "Marketing Theory with a Strategic Orientation", *Journal of Marketing*, Fall, pp. 79–89.

Fruhan, W. E. (1972) "Pyrrhic Victories in Fights for Market Share", *Harvard Business Review*, **50**, Sep/Oct, pp. 100–107.**

Goold, M. (1981a) "How 'Dogs' Can be Given More Bite", *Financial Times*, November 13, p. 16.**

Goold, M. (1981b) "Why Dicey Definitions are so Dangerous", *Financial Times*, November 16, p. 16.**

Goold, M. and Campbell, A. (1987) *Strategies and Styles*, Oxford, Blackwell.**

Hambrick, D. C., MacMillan, I. C. and Day, D. L. (1982) "Strategic Attributes and Performance in the BCG Matrix: A PIMS based Analysis of Industrial Product Businesses", *Academy of Management Journal*, September.**

Haspslagh, P. (1982) "Portfolio Planning: Uses and Limits", *Harvard Business Review*, Jan/Feb, pp. 58–73.

Hax, A. and Majluf, N. S. (1983) "The Use of the Growth Share Matrix in Strategic Planning", *Interfaces*, **13**, No. 1, February, pp. 46–60.

Hedley, B. (1977) "Strategy and the Business Portfolio", *Long Range Planning*, **10**, February, pp. 9–15.

Henderson, B. (1968) "Perspectives on Experience", Unpublished Monograph, BCG.

Henderson, B. (1970) "The Product Portfolio", *BCG Perspectives*, **66**.

Henderson, B. (1973) "The Experience Curve Reviewed: IV The Growth Share Matrix or The Product Portfolio", *BCG Perspectives*, **135**.

Henderson, B. (1980) "Strategic and Natural Competition", *BCG Perspectives*, **231**.

Hewitt, G. (1988) Unpublished Lecture, Barclays Bank Seminar, July 14.

Downloaded by [] at 01:07 16 May 2015

Kiechel, W. III (1979) "Playing by the Rules of the Corporate Strategy Game", *Fortune*, September, pp. 110–115.
Kiechel, W. III (1988) "Corporate Strategy for the 1990's", *Fortune*, February, pp. 34–42.
Kotler, P. (1984) *Marketing Management*, Englewood Cliffs, NJ, Prentice Hall.**
Lockridge, R. K. (1981) "Strategy in the 1980's", *BCG Perspectives*, **241**.
Lorenz, C. (1981) "Why Boston Recanted its Doctrine of Market Leadership", *Financial Times*, November 20, p. 9.
Lorrange, P. (1975) "Divisional Planning: Setting Effective Direction", *Sloan Management Review*, Fall, pp. 77–91.**
Majaro, S. (1977) "Market Share: Deception or Diagnosis", *Marketing*, March, pp. 43–47.**
Marsh, P., Barwise, P., Thomas, K. and Wensley, R. (1988) *Managing Strategic Investment Decisions in Large Diversified Companies*, CBS Report: London Business School.**
Porter, M. (1980) *Competitive Strategy: Techniques for Analysing Industries and Competitors*, New York, The Free Press.
Rogers, E. M. (1983) *The Diffusion of Innovations*, New York, The Free Press.
Schloeffler, S., Buzzell, R. D. and Heaney, D. F. (1974) "Impact of Strategic Planning on Profit Performance", *Harvard Business Review*, **52**, Mar/April, pp. 137–145.
Tilles, S. (1966) "Strategies for Allocating Funds", *Harvard Business Review*, **44**, Jan/Feb, pp. 72–80.
Townsend, R. (1970) *Up the Organization*, London, Joseph.
Wensley, R. (1981) "Strategic Marketing: Betas, Boxes or Basics", *Journal of Marketing*, Summer, pp. 173–183.**
Wilson, A. and Atkin, B. (1976) "Exorcising the Ghosts in Marketing", *Harvard Business Review*, **54**, Sep/Oct, pp. 117–127.
Wind, Y. and Mahajan, V. J. (1981) "Designing Product and Business Portfolios", *Harvard Business Review*, **59**, Jan/Feb, pp. 155–165.
Wind, Y. and Claycamp, H. J. (1976) "Planning Product Line Strategy—a Matrix Approach", *Journal of Marketing*, **40**, No. 1, January, pp. 2–9.**
Wind, Y., Mahajan, V. J. and Swire, D. J. (1981) "Standardised Portfolio Models: An Empirical Comparison of Business Classifications", *Proceedings of MSI Conference on Analytical Approaches to Product and Marketing Planning*, October, pp. 312–320.**
Wolman, C. (1988) "How Group can Challenge Management's Strategic Choices", *Financial Times*, March 8.**

** Indicates this article is only referred to in the Appendix.

Downloaded by [] at 01:07 16 May 2015